HR ANALYTICS DASHBOARD

Comprehensive Project Report

Built with Microsoft Power BI

Subject: Business Intelligence & Data Analytics

1. Introduction

Employee attrition is one of the most critical challenges faced by modern organizations. High attrition rates lead to increased recruitment costs, loss of institutional knowledge, productivity dips, and reduced team morale. Understanding the underlying causes of attrition is essential for HR departments to take proactive and data-driven decisions.

This project presents an HR Analytics Dashboard developed using Microsoft Power BI to analyze employee attrition patterns across a mid-sized organization. The dashboard enables HR managers and business leaders to interactively explore workforce data, identify at-risk segments, and design targeted retention strategies.

2. Data Source

2.1 Dataset Overview

The dataset used for this project is the IBM HR Analytics Employee Attrition & Performance dataset, which is publicly available on Kaggle. This is one of the most widely used HR datasets for machine learning and analytics projects.

Attribute	Details
Dataset Name	IBM HR Analytics Employee Attrition & Performance
Source	Kaggle (publicly available, open-source)
File Format	CSV (Comma-Separated Values)
Total Rows (Employees)	1,470
Total Columns (Features)	35
Time Period	Cross-sectional (single snapshot)
Tool Used	Microsoft Power BI Desktop

2.2 Key Columns in the Dataset

The 35 columns cover three major categories:

Category	Columns Included
Employee Demographics	Age, Gender, Education, EducationField, MaritalStatus, DistanceFromHome
Job Information	Department, JobRole, JobLevel, BusinessTravel, OverTime, YearsAtCompany, YearsInCurrentRole
Satisfaction & Performance	JobSatisfaction, EnvironmentSatisfaction, WorkLifeBalance, PerformanceRating, MonthlyIncome

Attrition Target Variable	Attrition (Yes/No) — the main column we are analyzing

3. Problem Statement

The organization faces a significant employee attrition challenge with 237 employees having left out of 1,470, resulting in a 16.12% attrition rate — nearly double the industry benchmark of 8–10%. The core questions driving this project are:

- Which departments have the highest attrition, and why?
- Which age groups are most likely to leave the organization?
- Does gender play a role in attrition within specific age brackets?
- Which education backgrounds are associated with higher attrition?
- How does job satisfaction vary across different job roles?
- What actionable steps can HR take to reduce attrition?

The goal is to convert raw HR data into meaningful visual insights that enable management to move from reactive hiring to proactive retention.

4. Data Cleaning & Transformation in Power Query Editor

Before building the dashboard, the raw CSV data was imported into Power BI and thoroughly cleaned using the Power Query Editor. The following transformations were performed:

4.1 Data Import

- Imported the CSV file using the Get Data > Text/CSV option in Power BI Desktop.
- Promoted the first row as headers using the Use First Row as Headers option.
- Auto-detected column data types were reviewed and corrected manually.

4.2 Data Type Corrections

- Changed Age, MonthlyIncome, YearsAtCompany, etc. from Text to Whole Number type.
- Ensured the Attrition column was treated as a Text type (Yes/No values).
- JobSatisfaction (1–4 scale) was kept as Whole Number for aggregation.

4.3 Creating Calculated Columns (DAX)

- Attrition Count: Created a DAX measure to count employees where Attrition = 'Yes' for KPI cards.
- Attrition Rate %: Calculated as (Attrition Count / Total Employees) * 100.

- CF_Age Band: A custom conditional column was added to group employees into age bands: Under 25, 25–34, 35–44, 45–54, Over 55.

Age Band Label	Condition Applied
Under 25	Age < 25
25–34	Age >= 25 AND Age < 35
35–44	Age >= 35 AND Age < 45
45–54	Age >= 45 AND Age < 55
Over 55	Age >= 55

4.4 Removing Irrelevant Columns

- Columns like EmployeeCount, Over18, and StandardHours were removed as they had constant values across all rows and added no analytical value.

4.5 Handling Null & Duplicate Values

- Checked for null/blank values using the Column Quality feature — no nulls were found in the IBM HR dataset.
- Removed duplicate rows (none were found, confirming data integrity).

4.6 Filtering & Sorting

- Applied slicers for Education Field (Human Resources, Life Sciences, Marketing, Medical, Other, Technical Degree) to enable dynamic dashboard filtering.
- Data was sorted by Department and Age Band for consistent charting.

5. Dashboard Components & Graph-by-Graph Explanation

5.1 KPI Summary Cards (Top Row)

Five large KPI cards are displayed prominently at the top of the dashboard, giving an instant snapshot of the organization's workforce health:

Overall Employees	**1,470**
Attrition	**237**
Attrition Rate	**16.12%**

Active Employees	**1,233**
Average Age	**37 years**

These cards use DAX COUNTROWS, CALCULATE, and DIVIDE functions to dynamically recalculate based on any active slicer selection (e.g., filtering by Education Field).

5.2 Department-wise Attrition (Donut Chart)

Chart Type: Donut / Pie Chart

This chart shows the breakdown of the 237 departed employees by department:

Department	Attrition Count	% Share
R&D (Research & Development)	133 employees left	56.12%
Sales	92 employees left	38.82%
HR (Human Resources)	12 employees left	5.06%

Insight: R&D and Sales account for over 94% of all attrition. This indicates systemic issues in these two departments — likely related to work pressure, limited career growth, or uncompetitive compensation packages.

5.3 Number of Employees by Age Group (Stacked Bar Chart)

Chart Type: Clustered/Stacked Bar Chart — Gender Split

This chart shows the total active employee headcount segmented by age band and gender:

Age Band	Female Count	Male Count
Under 25	Female: ~small	Male: ~small
25–34	Female: 217	Male: 337
35–44	Female: 196	Male: 309
45–54	Female: 113	Male: 132
Over 55	Small values	Small values

Insight: The 25–34 age group is the largest segment (554 employees), making them both the most valuable talent pool and the highest attrition risk group. Males consistently outnumber females across all bands.

5.4 Job Satisfaction Rating Table

Chart Type: Matrix / Cross-tab Table

This matrix shows each Job Role rated on a 1–4 scale (1 = Very Dissatisfied, 4 = Very Satisfied) with total employee counts:

Job Role	Total	Observation
Sales Executive	326	Highest headcount — highest attrition risk
Research Scientist	292	Strong satisfaction at level 3 & 4
Laboratory Technician	259	High dissatisfaction at levels 1 & 2
Manufacturing Director	145	Moderate across all levels
Healthcare Representative	131	Balanced distribution
Manager	102	Relatively satisfied
Research Director	80	Small, mostly satisfied
Sales Representative	83	Moderate dissatisfaction
Human Resources	52	Smallest role group

Insight: Laboratory Technicians and Sales Executives show significant low satisfaction scores (1 & 2), which directly correlates with their higher attrition rates.

5.5 Education Field-wise Attrition (Horizontal Bar Chart)

Chart Type: Horizontal Bar Chart

This chart ranks education fields by total attrition count:

Education Field	Attrition Count
Life Sciences	89 employees left
Medical	63 employees left
Marketing	35 employees left
Technical Degree	32 employees left
Other	11 employees left
Human Resources	7 employees left

Insight: Life Sciences graduates have the highest attrition (89), nearly 37% of all departures. This may reflect that Life Sciences talent is in high demand externally and more likely to receive competitive offers from competitor organizations.

5.6 Attrition Rate by Gender for Different Age Groups (Donut Charts)

Chart Type: 5 Donut Charts (one per age band) — Male vs Female split

These five donut charts show gender-based attrition within each age band:

Age Band	Total Attrition	Male	Female
Under 25	38 total	Male: 20 (52.63%)	Female: 18 (47.37%)
25–34	163 total	Male: 69 (61.61%)	Female: 43 (38.39%)
35–44	89 total	Male: 37 (72.55%)	Female: 14 (27.45%)
45–54	25 total	Male: 16 (64%)	Female: 9 (36%)
Over 55	11 total	Male: 8 (72.73%)	Female: 3 (27.27%)

Insight: Male employees account for a disproportionately higher share of attrition across all age groups, especially in the 35–44 and Over 55 bands. The 25–34 group has the highest absolute attrition (163 employees), making it the most critical segment.

6. Key Insights from the Dashboard

#	Insight
1	16.12% overall attrition rate — significantly above the 8–10% industry standard.
2	R&D department loses the most people (56.12%), suggesting career growth bottlenecks.
3	The 25–34 age group has the highest absolute attrition (163 out of 237 total).
4	Male employees leave more than female employees across all age bands.
5	Life Sciences graduates are the most likely to leave the organization.
6	Sales Executives and Laboratory Technicians have the lowest job satisfaction scores.
7	HR department has the lowest attrition, suggesting a more stable and supportive environment.
8	Attrition among employees Over 55 (72.73% male) likely includes voluntary early retirement.

7. Suggested Solutions & HR Recommendations

7.1 Targeted Retention Programs for R&D & Sales

- Introduce structured career pathing frameworks with clear promotion timelines for R&D employees.
- Create performance-linked incentive schemes specifically for the Sales department.
- Launch quarterly stay interviews in high-attrition teams to identify grievances early.

7.2 Early Engagement of Young Employees (25–34 Age Group)

- Introduce mentoring programs pairing junior employees (25–34) with senior leaders.
- Offer learning & development stipends and sponsored certifications to boost retention.
- Ensure competitive compensation benchmarked against market rates for this age cohort.

7.3 Addressing Male Attrition

- Conduct targeted surveys for male employees in R&D and Sales to uncover unique stressors.
- Review workload distribution and overtime policies that may disproportionately affect male employees.

7.4 Improving Job Satisfaction for Laboratory Technicians & Sales Representatives

- Introduce role-specific rewards and recognition programs for Laboratory Technicians.
- Redesign job structures to reduce monotony and include rotation programs.
- Train managers in Sales teams on motivational leadership techniques.

7.5 Retaining Life Sciences Talent

- Offer retention bonuses with 2–3 year cliff vesting schedules for Life Sciences graduates.
- Create innovation labs or research sabbaticals to keep Life Sciences talent intellectually engaged.

8. Conclusion & Summary of Key Findings

This HR Analytics Dashboard project demonstrates the power of data visualization in transforming raw HR records into actionable business intelligence. Using Microsoft Power BI, we analyzed 1,470 employee records across 35 dimensions to produce a comprehensive picture of the organization's attrition landscape.

Summary of Key Findings

Finding	Detail
Total Employees Analyzed	1,470 employees across R&D, Sales, and HR departments
Overall Attrition Rate	16.12% — significantly above industry average
Highest Risk Department	R&D (56.12% of all attrition)
Highest Risk Age Group	25–34 (163 departures — 68.8% of all attrition)
Gender Trend	Male employees leave at higher rates across all age bands

Education Risk	Life Sciences graduates account for the highest attrition count (89)
Satisfaction Gap	Sales Executives & Laboratory Technicians most dissatisfied
Most Stable Group	HR department employees — lowest attrition (5.06%)

The dashboard's interactive slicers (Education Field) allow HR managers to drill down into specific talent segments without needing technical skills, making it a practical decision-support tool for everyday HR operations.

By acting on these insights — particularly by focusing retention efforts on R&D talent aged 25–34 with Life Sciences backgrounds — the organization has the potential to reduce attrition by an estimated 30–40%, resulting in significant cost savings in recruitment, onboarding, and productivity losses.

End of Report

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